

PRACTICE PROBLEM SET 2
MTH 210
STATISTICAL COMPUTING

1. Suppose U_1 , U_2 and U_3 are three independent Weibull random variables, where $U_1 \sim \text{WE}(3, \lambda_1)$, $U_2 \sim \text{WE}(3, \lambda_2)$ and $U_3 \sim \text{WE}(3, \lambda_2)$, respectively. Suppose $X = \min\{U_1, U_3\}$ and $Y = \min\{U_2, U_3\}$. If you have a random sample $(X_1, Y_1), \dots, (X_n, Y_n)$ from (X, Y) , then find the MLEs of λ_1 , λ_2 and λ_3 by Newton-Raphson method also by using EM algorithm.
2. Suppose U_1 , U_2 and U_3 are three independent Weibull random variables, where $U_1 \sim \text{WE}(\alpha, \lambda_1)$, $U_2 \sim \text{WE}(\alpha, \lambda_2)$ and $U_3 \sim \text{WE}(\alpha, \lambda_2)$, respectively. Suppose $X = \min\{U_1, U_3\}$ and $Y = \min\{U_2, U_3\}$. If you have a random sample $(X_1, Y_1), \dots, (X_n, Y_n)$ from (X, Y) , then find the MLEs of α , λ_1 , λ_2 and λ_3 by Newton-Raphson method also by using EM algorithm.
3. Let $X_1, \dots, X_n$ be a random sample from the following PDF

$$f(x; \alpha, \lambda) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}; \quad x > 0.$$

You need to find the MLEs of α and λ . So that you can obtain the MLEs of α and λ by solving one non-linear equation only.

4. In Problem 3, find the MLEs of α and λ based on one dimensional Newton-Raphson method.
5. In Problem 3, find the MLEs of α and λ based on bisection method.
6. Find the MLE of α and μ for Pareto distribution with PDF

$$f(x) = \frac{\alpha \mu^\alpha}{x^{\alpha+1}}; \quad x \geq \mu, \mu > 0, \alpha > 0.$$

Do you need numerical procedure to compute MLEs here?

7. Find the MLEs of α and β for a Beta(α, β) distribution. Do you need numerical procedure to compute MLEs here?
8. Find the MLE of μ from a random sample of a Cauchy distribution

$$f(x) = \frac{1}{\pi(1 + (x - \mu)^2)}; \quad -\infty < \mu < \infty, \quad -\infty < x < \infty.$$

Do you need numerical procedure to compute MLEs here?